

Reducing Turbulence-Induced Outages in a Deployed Terrestrial Free-Space Optical Communication Link via Interleaving

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Abstract We present an experimental study of data interleaving for terrestrial free-space optical communication over a 4.6 km urban testbed. Results demonstrate a two-order-of-magnitude reduction in outage probability. A dependency between measured turbulence strength, interleaver length, and achievable data rate is revealed, enabling robust system design. © 2026 The Author(s)

Introduction

Free-space optical (FSO) communication can enable high-data-rate wireless communication by leveraging technologies used in coherent optical fibre transmission. Although more commonly studied for non-terrestrial links^[1,2], high-capacity terrestrial links are starting to be demonstrated experimentally in recent years^[3,4]. Terrestrial use cases include emergency networks, cellular backhaul, and security applications such as quantum key distribution^[5], which require high reliability.

One of the main challenges in FSO is atmospheric turbulence, which causes significant fluctuations in the system's received optical power and introduces random fluctuations in irradiance. Hence, the system's signal-to-noise ratio (SNR), and therefore its data rate, changes over time. For a terrestrial link, these fluctuations can be more severe than in satellite systems because the beam travels farther through the dense part of the atmosphere. This affects the reliability of terrestrial

FSO links, leading to high outage probability and preventing widespread deployment^[5].

Data interleaving has been proposed as an effective mitigation technique^[6], redistributing burst errors caused by turbulence across multiple code-words to improve error-correction performance. These interleavers are often used in conjunction with other link-layer solutions, such as automatic repeat request and its variants^[7], which is outside of the scope of this paper. Interleaving has been investigated extensively for satellite-based FSO systems^[8,9]. Although experimental works on terrestrial links do exist^[10,11], they do not analyse how outage probability is affected by interleavers under realistic urban atmospheric conditions.

In this work, we address this gap by analysing interleaving in a deployed terrestrial FSO system. Using the Reid Photonloop testbed^[3], a 4.6 km urban link, we quantify how interleaver length influences outage probability under varying turbulence conditions. Furthermore, we show a clear relation-

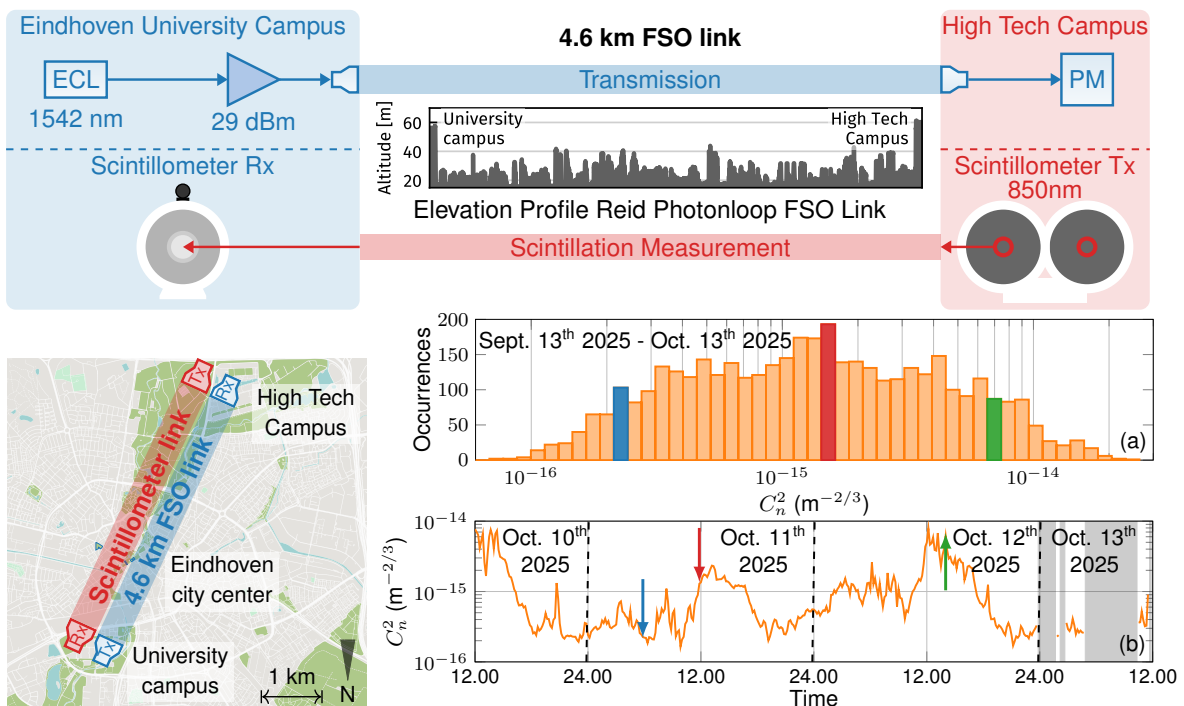


Fig. 1: Experimental setup for power and scintillation measurements over 4.6 km for a free-space optical link. The insets show (a) the distribution of measured C_n^2 from September 13th to October 13th 2025 and (b) the C_n^2 measurements from October 10th 12.00 to October 13th 12.00. For insufficient received power the background is grey. The blue, red and green correspond to the investigated weak, moderate and strong turbulence cases.

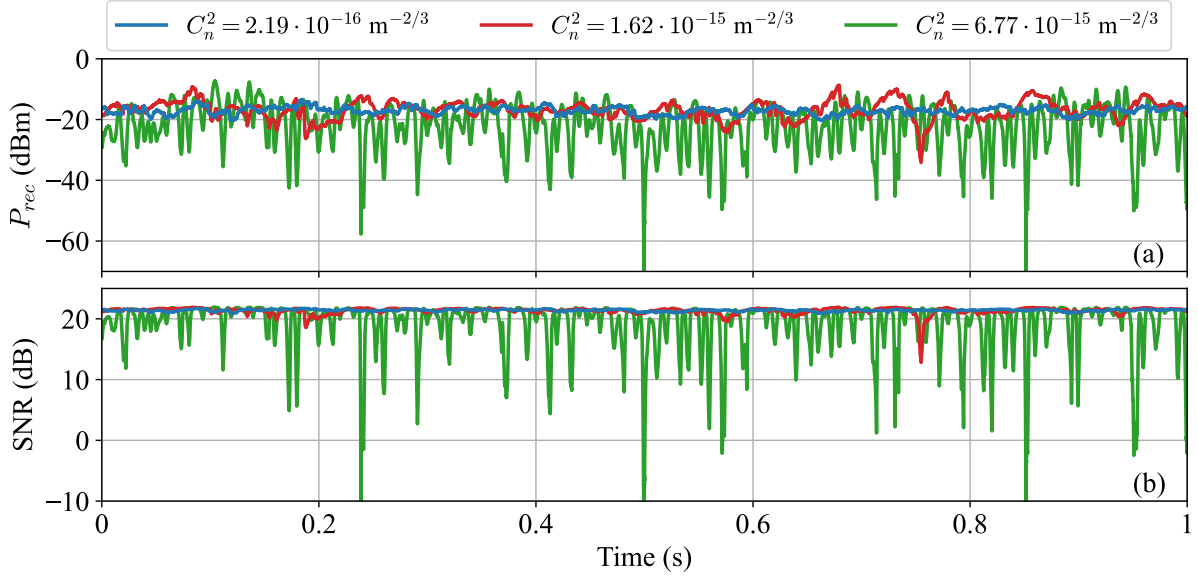


Fig. 2: (a): Received optical power, P_{rec} (dBm) vs time, (b) SNR (dB) vs time for 3 different turbulence cases.

ship among turbulence strength, interleaver length, and data rate, providing practical design insights for reliable terrestrial FSO deployment.

Experimental Setup

Our terrestrial FSO setup, as shown in Fig. 1, spans 4.6 km over the city of Eindhoven, from the Eindhoven University of Technology Campus to the High Tech Campus (HTC). To measure the received optical power P_{rec} at the FSO link receiver, we use an external cavity laser (ECL) at 1542 nm, which is amplified by an erbium-doped fiber amplifier (EDFA) to 29 dBm. The light is transmitted across the city via fibre-coupled optical terminals designed by Aircision. On the HTC side, we measure the received power P_{rec} using a high-speed power meter at 10 μ s intervals. The power measurements ran from October 10th to 13th 2025.

In parallel with our power measurements, we measure turbulence strength using a co-located Scintec BLS900 Neo Dual-Disk Design scintillometer. This measurement system characterises the optical turbulence in the FSO channel using the refractive index structure parameter C_n^2 , with an averaging time of 10 minutes^[12]. Although traversing the FSO link in opposite directions, the scintillometer and power meter measurements do correlate. Fig. 2 shows that for higher measured C_n^2 , the P_{rec} fluctuations on the FSO receiver are more severe. This correlation was observed across our measurements. The scintillation measurements are from September 13th to October 13th 2025.

In the insets in Fig. 1 (a), the distribution of C_n^2 for the period of one month and (b) C_n^2 measured in the same period as the optical power measurements are shown. For 4.2% of the scintillation measurements, the received optical power was below the threshold for the scintillometer receiver to measure C_n^2 . These measurements are not taken

into account in inset (a) and are shown with a grey background in inset (b). The distribution of C_n^2 is centered around $1.5 \cdot 10^{-15} \text{ m}^{-2/3}$, with two more peaks around $5 \cdot 10^{-16} \text{ m}^{-2/3}$ and $4 \cdot 10^{-15} \text{ m}^{-2/3}$, caused by a higher level of atmospheric turbulence during the day, and lower at night, as shown by the diurnal pattern indicated in inset (b).

To investigate the effect of bit interleaving on the outage probability P_{out} , the probability that the decoded code rate is lower than the operational code rate, we take a one second sample from our power meter results corresponding to weak (blue, $C_n^2 = 2.19 \cdot 10^{-16} \text{ m}^{-2/3}$), moderate (red, $C_n^2 = 1.62 \cdot 10^{-15} \text{ m}^{-2/3}$, and strong (green, $C_n^2 = 6.77 \cdot 10^{-15} \text{ m}^{-2/3}$) observed turbulence. These samples were chosen based on the distribution of C_n^2 shown in Fig. 1(a), picking samples in the middle, and toward the tail ends of the distribution. The received power P_{rec} for these traces is shown in Fig. 2 (a). We assume the following for our system: We use an EDFA with a noise figure of 4.4 dB to amplify the received signal to 0 dBm. To calculate the ASE noise generated by the EDFA, we use eq. (54) from^[13]. The symbol rate B of our system is 50 GBaud, using 256-QAM. The SNR_{B2B} we measured of our back-to-back system

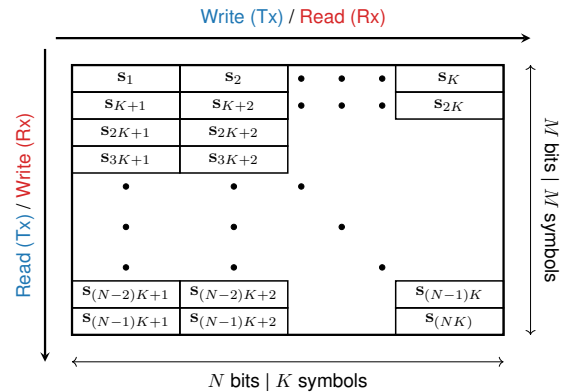


Fig. 3: Overview of a symbol-wise block (de)interleaver.

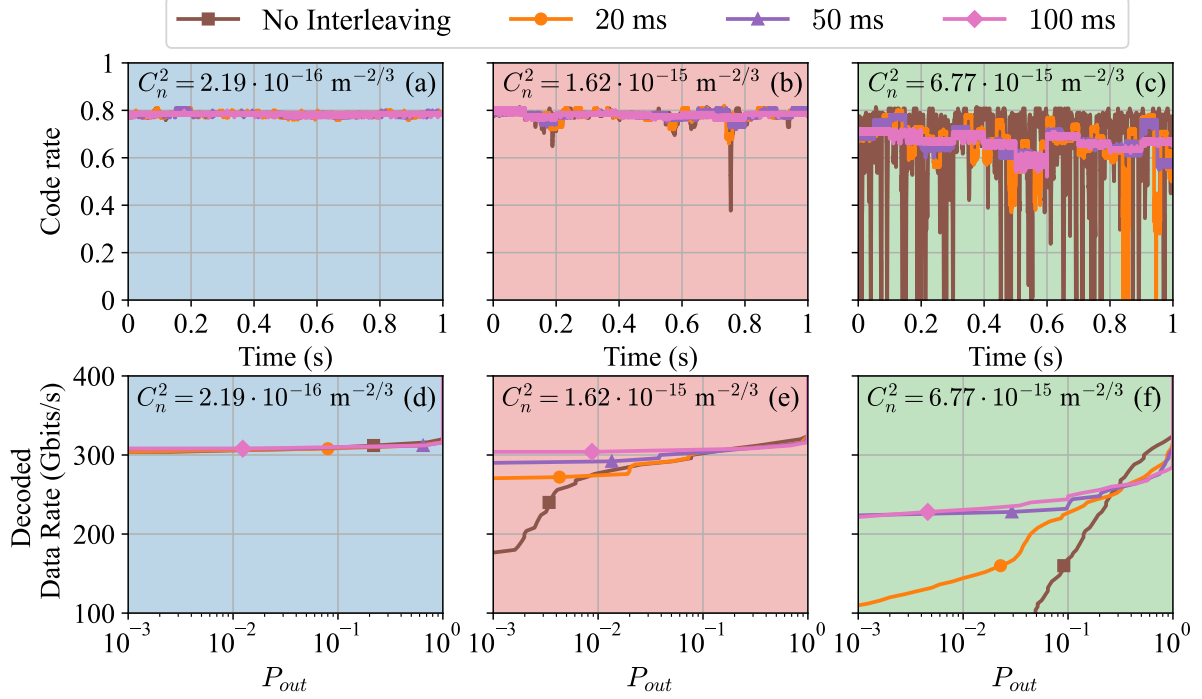


Fig. 4: (a, b, c) Code rate vs time, (d, e, f) Throughput vs P_{out} for four different interleaver lengths. (a,d): weak turbulence, (b,e): moderate turbulence, (c,f): strong turbulence.

is 22 dB. The total SNR, as shown in Fig. 2 (b), is $SNR = \frac{1}{1/SNR_{B2B} + 1/SNR_{EDFA}}$, where SNR_{EDFA} is the SNR of a shot-noise-limited signal amplified by the receiver EDFA. Both P_{rec} and the SNR fluctuate more for increasing turbulence.

Decoding with Interleaver

For the error correction, we use punctured low-density parity-check (LDPC) codes from the DVB-S2 standard^[14], which have an error correction blocklength N of 64800 bits. The minimum rate for the DVB-S2 standard is $\frac{1}{4}$; hence, whenever a lower rate is required, decoding fails and there is no throughput. The block (de)interleaver used in the simulation is shown in Fig. 3. The size of the interleaver depends on N , the length of the interleaver in seconds t , and the symbol rate B . The interleaver has K columns, where $K = \frac{N}{m}$ and m is the number of bits per symbol s_i , and $M = \frac{tB}{N}$ rows. On the transmitter side, we write our symbol stream into the interleaver row by row. After filling the interleaver matrix, we read out the symbols column by column and transmit them over the channel in this order. At the receiver side, we do the exact opposite: write to the deinterleaver column by column and read out row by row, returning to the original symbol order. Any burst errors caused by fast fading are now spread over multiple codewords. However, the interleaver introduces a latency equal to $2t$, as the interleaver and deinterleaver need to be completely filled before transmission and decoding can start, respectively. For the error correction simulation, we simulate per 10 μ s time instance down to an outage prob-

ability P_{out} of 10^{-3} . In Fig. 4, we show the code rate vs time and the throughput vs P_{out} for all 3 investigated turbulence scenarios and different interleaver lengths. In the low-turbulence case, throughput is consistent over time, and interleaving does not significantly affect P_{out} . In the moderate case, interleaving allows for up to a 70% higher decoded data rate compared to no interleaving when $P_{out} < 10^{-2}$ due to the deep fade at 0.75 s. The 20 ms interleaver significantly reduces the outage probability, with longer interleaver lengths yielding incremental gains in decoded data rate. For the high-turbulence case, an interleaver of at least 50 ms is required to get $P_{out} < 10^{-2}$ at decoded data rates above 200 Gb/s. These results show that as C_n^2 increases, the channel becomes less stable, and longer interleavers are required for high data rates at low P_{out} , at the cost of higher latency.

Conclusion

This work experimentally characterised the distribution of C_n^2 over a 4.6 km terrestrial FSO link and quantified its impact on outage probability. We show that interleaving is a decisive factor in mitigating turbulence-induced outages, with optimal interleaver length strongly dependent on turbulence strength. In particular, increasing C_n^2 necessitates longer interleavers to sustain high data rates at low outage probabilities, in our case $P_{out} < 10^{-2}$ for data rates ≥ 200 Gb/s, at the cost of higher latency. These results establish a direct link between atmospheric conditions and interleaver design, providing a foundation for adaptive interleaving strategies in practical terrestrial FSO systems.

Acknowledgements

Supported by the Dutch Research Council (NWO) TTW-Perspectief Optical Wireless Super-highways: Free photons (at home and in space): FREE P19-13, the Dutch Ministry of Economic Affairs and Climate Policy (EZK) via the Photon-Delta National Growth Fund Programme on Photonics, and European Innovation Council Transition project CombTools (G.A. 101136978). We thank Aircision B.V., particularly Nourdin Kaai, Luis Pellicer Collado, and Roland Blok, for their support of the Reid Photonloop FSO testbed, and Keysight Technologies for providing the high-speed power meter.

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